# Exponential Functions and Their Graphs

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Evaluating and describing

Let $f(x) = 3 \cdot 2^x$. Evaluate:

**a** $f(0) = 3$ **b** $f(3) = 24$ **c** $f(-2) = \frac{3}{4}$

(a) $= 3$; (b) $= 24$; (c) $= \frac{3}{4}$

**2** [Foundational] • Target: Evaluating and describing

For $y = 5^x$, state:

**a** the domain All real numbers.

**b** the range $y > 0$.

**c** the horizontal asymptote $y = 0$.

**d** the $y$-intercept $(0, 1)$.

(a) All real numbers.; (b) $y > 0$.; (c) $y = 0$.; (d) $(0, 1)$.

**3** [Foundational] • Target: Evaluating and describing

Describe how the graph of $g(x) = 2^{x-3} + 1$ is obtained from the graph of $y = 2^x$, and state the asymptote of $g$.

Shift right 3 units and up 1 unit. Horizontal asymptote: $y = 1$.

**4** [Foundational] • Target: Identifying from graphs and applications

The graph below passes through $(0, 1)$, $(1, 2)$, and $(2, 4)$. Write the equation of the exponential function shown.

$y = 2^x$.

**5** [Foundational] • Target: Identifying from graphs and applications

An investment of 1000 riyals grows with continuous compounding at an annual rate of 6%: $A(t) = 1000e^{0.06t}$. Find the value after 5 years, to the nearest riyal.

$A(5) = 1000e^{0.3} \approx 1350$ riyals.

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**6** [Foundational] • Target: Identifying from graphs and applications

Determine whether each function represents growth or decay:

**a** $y = 4(1.08)^x$ Growth.

**b** $y = 7(0.85)^x$ Decay.

**c** $y = 2e^{-0.3x}$ Decay.

**d** $y = \left(\frac{5}{4}\right)^x$ Growth.

(a) Growth.; (b) Decay.; (c) Decay.; (d) Growth.

**7** [Foundational] • Target: Compound interest

2000 riyals is invested at 5% annual interest, compounded monthly, using $A = P(1 + \frac{r}{n})^{nt}$.

**a** Find the value after 3 years, to the nearest riyal. $A = 2000\left(1 + \frac{0.05}{12}\right)^{36} \approx 2000(1.1616) \approx 2323$ riyals.

**b** Find the value after 3 years if compounded continuously instead ($A = Pe^{rt}$), to the nearest riyal. $A = 2000e^{0.15} \approx 2000(1.1618) \approx 2324$ riyals.

(a) $A = 2000\left(1 + \frac{0.05}{12}\right)^{36} \approx 2000(1.1616) \approx 2323$ riyals.; (b) $A = 2000e^{0.15} \approx 2000(1.1618) \approx 2324$ riyals.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Compound interest

How much must be invested now at 4% annual interest compounded annually to have 10,000 riyals in 8 years? Round to the nearest riyal.

$P = \frac{10000}{(1.04)^8} \approx \frac{10000}{1.3686} \approx 7308$ riyals.

**9** [Intermediate] • Target: Comparing exponential growth rates

Two investments start at 500 riyals: one grows at 4% per year, the other at 7% per year. Find the value of each after 10 years, to the nearest riyal, and their difference.

4%: $500(1.04)^{10} \approx 740$ riyals. 7%: $500(1.07)^{10} \approx 984$ riyals. Difference $\approx 244$ riyals.

**10** [Intermediate] • Target: Comparing exponential growth rates

Rank the following functions from slowest to fastest growth as $x \to \infty$: $y = x^{10}$, $y = 2^x$, $y = 1.01^x$.

Any exponential with base $> 1$ eventually outgrows any polynomial, no matter how slowly it starts. So even though $1.01^x$ grows very slowly at first, it eventually overtakes $x^{10}$. Long-run ranking (slowest to fastest): $x^{10} < 1.01^x < 2^x$.

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**11** [Intermediate] • Target: Comparing growth and decay graphically

The graphs of $y = 2^x$ (growth) and $y = \left(\frac{1}{2}\right)^x$ (decay) are shown together. Use the graph to describe the symmetry relating the two curves.

The two curves are mirror images of each other across the $y$-axis: $\left(\frac{1}{2}\right)^x = 2^{-x}$, so reflecting $y = 2^x$ over the $y$-axis (replacing $x$ with $-x$) produces the decay curve.

**12** [Intermediate] • Target: The natural exponential base e

Explain what distinguishes $e \approx 2.71828$ from other exponential bases, in terms of the slope of $y = e^x$ at $x = 0$.

The number $e$ is the unique base for which the exponential function $y = b^x$ has a tangent line slope of EXACTLY $1$ at $x = 0$ — i.e., the function's instantaneous rate of change equals its own value at every point, a property no other base shares.

**13** [Intermediate] • Target: The natural exponential base e

Evaluate $e^0$, $e^1$ (to 3 decimal places), and simplify $e^{\ln 5}$.

$e^0 = 1$. $e^1 \approx 2.718$. $e^{\ln 5} = 5$ (since $e^x$ and $\ln x$ are inverse functions).

**14** [Intermediate] • Target: Piecewise-exponential applications

A machine depreciates in value according to $V(t) = 15000(0.88)^t$ (riyals, years). Find its value after 5 years, and the year in which its value first drops below 5000 riyals (to the nearest year).

$V(5) = 15000(0.88)^5 \approx 15000(0.5277) \approx 7916$ riyals. Setting $15000(0.88)^t < 5000$: $(0.88)^t < 0.333$. Testing values: at $t = 9$, $(0.88)^9 \approx 0.316 < 0.333$; at $t = 8$, $(0.88)^8 \approx 0.360 > 0.333$. First drops below 5000 riyals in year $9$.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

State the horizontal asymptote and range of each transformed exponential function:

**a** $y = 3^x + 4$ Asymptote $y = 4$; range $y > 4$.

**b** $y = -2^x$ Asymptote $y = 0$; range $y < 0$ (reflected below the axis).

**c** $y = 5^x - 3$ Asymptote $y = -3$; range $y > -3$.

(a) Asymptote $y = 4$; range $y > 4$.; (b) Asymptote $y = 0$; range $y < 0$ (reflected below the axis).; (c) Asymptote $y = -3$; range $y > -3$.

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**16** [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Write the equation of the function obtained by taking $y = e^x$, reflecting it across the $y$-axis, then shifting it down 2 units.

Reflecting across the $y$-axis replaces $x$ with $-x$: $y = e^{-x}$. Shifting down 2: $y = e^{-x} - 2$.

**17** [Challenging] • Target: Comparing exponential functions numerically • Cross-Lesson: 3.01 The Unit Circle and Exact Values

For $f(x) = 3^x$, complete a short table of values at $x = -1, 0, 1, 2, 3$ and verify each value satisfies $f(x+1) = 3f(x)$.

$f(-1) = \frac{1}{3}$, $f(0) = 1$, $f(1) = 3$, $f(2) = 9$, $f(3) = 27$. Check: $f(0) = 3f(-1) = 3(1/3) = 1$ ✓; $f(1) = 3f(0) = 3$ ✓; $f(2) = 3f(1) = 9$ ✓; $f(3) = 3f(2) = 27$ ✓ — each step multiplies by the base $3$, confirming the defining property of exponential growth.

**18** [Challenging] • Target: Comparing growth and decay graphically • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the inverse of $f(x) = \frac{4x-1}{2x+3}$.

$y = \frac{4x-1}{2x+3} \Rightarrow y(2x+3) = 4x - 1 \Rightarrow x = \frac{-3x-1}{2(x-2)}$. Final Answer: $\frac{-3x-1}{2(x-2)}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

**19** [Challenging] • Target: The natural exponential base e • Cross-Lesson: 3.03 Sum and Difference Formulas

Solve $7^x = 100$ for $x$, expressed using logarithms.

Take $\ln$ of both sides: $x \ln 7 = \ln 100$, so $x = \frac{\ln 100}{\ln 7} \approx 2.367$. Final Answer: $x = \log_7(100) = \frac{\ln 100}{\ln 7}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

**20** [Challenging] • Target: Piecewise-exponential applications • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Find the 15th term and the sum of the first 20 terms of the arithmetic progression $4, 9, 14, 19, \ldots$.

Step 1: Identify progression parameters: First term $a_1 = 4$, common difference $d = 9 - 4 = 5$. Step 2: Compute $a_{15}$: $a_{15} = a_1 + (15-1)d = 4 + 14(5) = 4 + 70 = 74$. Step 3: Compute finite sum $S_{20}$: $S_{20} = \frac{20}{2}[2a_1 + (20-1)d] = 10[2(4) + 19(5)] = 10[8 + 95] = 10(103) = 1030$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Base e and Exponential Growth/Decay Curves

For $f(x) = a \cdot b^x$ with $a > 0$, under what condition does $f(x)$ represent exponential growth?

**a.** $b > 1$ ✓

**b.** $0 < b < 1$

**c.** $b < 0$

**d.** $b = 1$

Exponential growth occurs when the base $b > 1$. If $0 < b < 1$, it represents exponential decay.

**22** [Foundational] • Target: Base e and Exponential Growth/Decay Curves

Find the horizontal asymptote and $y$-intercept of $f(x) = 3 \cdot 2^x - 5$.

**a.** **Asymptote:** $y = -5$**,** $y$**-intercept:** $(0, -2)$ ✓

**b.** Asymptote: $y = 0$, $y$-intercept: $(0, 3)$

**c.** Asymptote: $y = -5$, $y$-intercept: $(0, -5)$

**d.** Asymptote: $x = 0$, $y$-intercept: $(0, -2)$

As $x \to -\infty$, $2^x \to 0$, so $y \to -5$. At $x = 0$: $f(0) = 3(1) - 5 = -2$.

**23** [Foundational] • Target: Base e and Exponential Growth/Decay Curves

Determine the range of $g(x) = -2 \cdot e^x + 4$.

**a.** $(-\infty, 4)$ ✓

**b.** $[4, \infty)$

**c.** $(4, \infty)$

**d.** $(-\infty, -2)$

Since $e^x > 0$, $-2e^x < 0$. Adding 4 gives $g(x) < 4$. The range is $(-\infty, 4)$.

**24** [Foundational] • Target: Transformations of Exponential Functions

Describe the transformations needed to obtain $g(x) = 2^{x-3} + 1$ from $f(x) = 2^x$.

**a.** **Shift right by 3 units and up by 1 unit** ✓

**b.** Shift left by 3 units and up by 1 unit

**c.** Shift right by 3 units and down by 1 unit

**d.** Vertical stretch by 3 and up by 1

Replacing $x$ by $x - 3$ shifts right 3; adding 1 shifts up 1.

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**25** [Foundational] • Target: Transformations of Exponential Functions

What is the domain of any function of the form $f(x) = a \cdot b^{x-h} + k$ $(b > 0, b \neq 1)$?

**a.** $(-\infty, \infty)$ ✓

**b.** $[0, \infty)$

**c.** $(k, \infty)$

**d.** $[h, \infty)$

Exponential functions are defined for every real number $x$, so the domain is always all real numbers $(-\infty, \infty)$.

**26** [Foundational] • Target: Transformations of Exponential Functions

Which transformation reflects the graph of $f(x) = e^x$ across the $y$-axis?

**a.** $g(x) = e^{-x}$ ✓

**b.** $g(x) = -e^x$

**c.** $g(x) = -e^{-x}$

**d.** $g(x) = e^x - 1$

Replacing $x$ with $-x$ reflects the graph across the $y$-axis.

**27** [Foundational] • Target: Compound Interest Formulas

State the formula for an initial principal $P$ compounded continuously at annual rate $r$ for $t$ years.

**a.** $A = Pe^{rt}$ ✓

**b.** $A = P(1 + r)^t$

**c.** $A = P(1 + \frac{r}{n})^{nt}$

**d.** $A = Prt$

Continuous compounding uses the natural base: $A = Pe^{rt}$.

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**28** [Foundational] • Target: Compound Interest Formulas

If 1000 dollars is invested at 6% compounded monthly, what is the balance after 5 years?

**a.** $1000\left(1+\frac{0.06}{12}\right)^{60} \approx 1348.85$ dollars ✓

**b.** 1300 dollars

**c.** 1349.86 dollars

**d.** 1338.23 dollars

$A = 1000(1+0.005)^{60} = 1000(1.005)^{60} \approx 1348.85$ dollars.

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## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Compound Interest Formulas

If 2000 dollars is invested at 5% compounded continuously, find the value after 10 years.

**a.** $2000e^{0.5} \approx 3297.44$ dollars ✓

**b.** 3000 dollars

**c.** 3257.79 dollars

**d.** 4000 dollars

$A = 2000e^{(0.05)(10)} = 2000e^{0.5} \approx 2000(1.64872) \approx 3297.44$ dollars.

**30** [Intermediate] • Target: The Number e as a Limit

State the limit definition of Euler's number $e$.

**a.** $e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$ ✓

**b.** $e = \lim_{n\to 0}\left(1+\frac{1}{n}\right)^n$

**c.** $e = \lim_{n\to\infty}(1+n)^{1/n}$

**d.** $e = \sum_{n=1}^{\infty}\frac{1}{n}$

Euler's number $e \approx 2.71828$ is defined by $\lim_{n\to\infty}(1+1/n)^n$.

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**31** [Intermediate] • Target: The Number e as a Limit

Evaluate $\lim_{n\to\infty}(1+\frac{3}{n})^n$.

**a.** $e^3$ ✓

**b.** $3e$

**c.** $e^{1/3}$

**d.** $3$

Using $\lim_{n\to\infty}(1+x/n)^n = e^x$ with $x = 3$ gives $e^3$.

**32** [Intermediate] • Target: The Number e as a Limit

Why is continuous compounding optimal compared to discrete compounding at the same nominal rate?

**a.** **As $n \to \infty$, the compounding factor $(1 + r/n)^{nt} \to e^{rt}$, which achieves the theoretical maximum return** ✓

**b.** It avoids paying bank fees

**c.** The interest rate increases

**d.** It doubles every year

Compounding frequency increases accumulation monotonically, reaching its mathematical supremum in the limit $n \to \infty$ with $e^{rt}$.

**33** [Intermediate] • Target: The Number e as a Limit

The graph of $g(x) = f(x - 4) - 5$ is obtained from $f(x)$ by:

**a.** Shifting 4 units left and 5 units up

**b.** Shifting 4 units right and 5 units down

**c.** Reflecting over the x-axis

**d.** **Shifting 4 units left and 5 units down** ✓

Replacing $x$ with $x + 4$ shifts horizontally by -4 (right if positive); adding -5 shifts vertically by -5.

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**34** [Intermediate] • Target: Evaluating and describing

The graph of $g(x) = f(x - 4) - 3$ is obtained from $f(x)$ by:

**a.** Reflecting over the x-axis

**b.** Shifting 4 units right and 3 units down

**c.** **Shifting 4 units left and 3 units down ✓**

**d.** Shifting 4 units left and 3 units up

Replacing $x$ with $x + 4$ shifts horizontally by -4 (right if positive); adding -3 shifts vertically by -3.

**35** [Intermediate] • Target: Identifying from graphs and applications

The graph of $g(x) = f(x - 5) - 4$ is obtained from $f(x)$ by:

**a.** Shifting 5 units left and 4 units up

**b.** Shifting 5 units right and 4 units down

**c.** Reflecting over the x-axis

**d.** **Shifting 5 units left and 4 units down ✓**

Replacing $x$ with $x + 5$ shifts horizontally by -5 (right if positive); adding -4 shifts vertically by -4.

**36** [Intermediate] • Target: Compound interest

The graph of $g(x) = f(x - 1) - 5$ is obtained from $f(x)$ by:

**a.** Shifting 1 units right and 5 units down

**b.** **Shifting 1 units left and 5 units down ✓**

**c.** Reflecting over the x-axis

**d.** Shifting 1 units left and 5 units up

Replacing $x$ with $x + 1$ shifts horizontally by -1 (right if positive); adding -5 shifts vertically by -5.

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**37** [Intermediate] • Target: Comparing exponential growth rates

The graph of $g(x) = f(x - 4) - 1$ is obtained from $f(x)$ by:

**a.** Reflecting over the x-axis

**b.** Shifting 4 units left and 1 units up

**c.** **Shifting 4 units left and 1 units down ✓**

**d.** Shifting 4 units right and 1 units down

Replacing $x$ with $x + 4$ shifts horizontally by -4 (right if positive); adding -1 shifts vertically by -1.

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Comparing growth and decay graphically • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

The graph of $g(x) = f(x - 2) - 4$ is obtained from $f(x)$ by:

**a.** Reflecting over the x-axis

**b.** Shifting 2 units left and 4 units up

**c.** **Shifting 2 units left and 4 units down ✓**

**d.** Shifting 2 units right and 4 units down

Replacing $x$ with $x + 2$ shifts horizontally by -2 (right if positive); adding -4 shifts vertically by -4.

**39** [Challenging] • Target: The natural exponential base e • Cross-Lesson: 3.03 Sum and Difference Formulas

The graph of $g(x) = f(x - 4) - 2$ is obtained from $f(x)$ by:

**a.** Shifting 4 units left and 2 units up

**b.** **Shifting 4 units left and 2 units down ✓**

**c.** Reflecting over the x-axis

**d.** Shifting 4 units right and 2 units down

Replacing $x$ with $x + 4$ shifts horizontally by -4 (right if positive); adding -2 shifts vertically by -2.

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**40** [Challenging] • Target: Piecewise-exponential applications • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

The graph of $g(x) = f(x + 1) - 5$ is obtained from $f(x)$ by:

a. Shifting 1 units left and 5 units down

b. **Shifting 1 units right and 5 units down ✓**

c. Shifting 1 units right and 5 units up

d. Reflecting over the x-axis

Replacing $x$ with $x - 1$ shifts horizontally by 1 (right if positive); adding -5 shifts vertically by -5.

**41** [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 4.01 Circles and Parabolas

The graph of $g(x) = f(x + 4) - 4$ is obtained from $f(x)$ by:

a. **Shifting 4 units right and 4 units down ✓**

b. Shifting 4 units left and 4 units down

c. Shifting 4 units right and 4 units up

d. Reflecting over the x-axis

Replacing $x$ with $x - 4$ shifts horizontally by 4 (right if positive); adding -4 shifts vertically by -4.

**42** [Challenging] • Target: Comparing exponential functions numerically • Cross-Lesson: 4.02 Ellipses

The graph of $g(x) = f(x + 2) - 5$ is obtained from $f(x)$ by:

a. **Shifting 2 units right and 5 units down ✓**

b. Shifting 2 units right and 5 units up

c. Reflecting over the x-axis

d. Shifting 2 units left and 5 units down

Replacing $x$ with $x - 2$ shifts horizontally by 2 (right if positive); adding -5 shifts vertically by -5.

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**43** [Challenging] • Target: Base e and Exponential Growth/Decay Curves • Cross-Lesson: 4.03 Hyperbolas

The graph of $g(x) = f(x + 1) - 4$ is obtained from $f(x)$ by:

**a.** Shifting 1 units left and 4 units down

**b.** Shifting 1 units right and 4 units up

**c.** **Shifting 1 units right and 4 units down ✓**

**d.** Reflecting over the x-axis

Replacing $x$ with $x - 1$ shifts horizontally by 1 (right if positive); adding -4 shifts vertically by -4.

**44** [Challenging] • Target: Transformations of Exponential Functions • Cross-Lesson: 5.01 Vectors: Components and Operations

The graph of $g(x) = f(x + 5) - 5$ is obtained from $f(x)$ by:

**a.** Shifting 5 units right and 5 units up

**b.** Shifting 5 units left and 5 units down

**c.** Reflecting over the x-axis

**d.** **Shifting 5 units right and 5 units down ✓**

Replacing $x$ with $x - 5$ shifts horizontally by 5 (right if positive); adding -5 shifts vertically by -5.

**45** [Challenging] • Target: Compound Interest Formulas • Cross-Lesson: 5.02 The Dot Product

The graph of $g(x) = f(x + 6) - 6$ is obtained from $f(x)$ by:

**a.** Reflecting over the x-axis

**b.** Shifting 6 units left and 6 units down

**c.** Shifting 6 units right and 6 units up

**d.** **Shifting 6 units right and 6 units down ✓**

Replacing $x$ with $x - 6$ shifts horizontally by 6 (right if positive); adding -6 shifts vertically by -6.

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